

Assignment 4

Week 4

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Problem 1. Find Eigenvalues using Colab

Find the eigenvectors and eigenvalues of σ_x using colab. You may use this code. But please ask yourself, which basis are we using when we express σ_x in the following form?

```
1 sigma = np.array([[0, 1], [1, 0]])
2 print(sigma)
3
4 eigenvalues, eigenvectors = np.linalg.eig(sigma)
5 print(eigenvalues)
6 print(eigenvectors)
```

Solution

After running the code in Colab, we find that the eigenvalues and eigenvectors of matrix σ_x are:

Eigenvalues	Eigenvectors	
	from code	by hand
1	$\begin{pmatrix} 0.7071 \\ 0.7071 \end{pmatrix}$	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
-1	$\begin{pmatrix} 0.7071 \\ -0.7071 \end{pmatrix}$	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

These results match our expectations from the theoretical calculation. Additionally, the basis we use to represent σ_x in matrix form is the standard computational basis $|0\rangle / |1\rangle$.

Problem 2. Construct Projection Operator

Now, use hand calculation, derive the projection operators to project to the eigenstates of σ_z .

Solution

To project the eigenstates of $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, we first identify its eigenvectors:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

These eigenvectors correspond to the eigenvalues $+1$ and -1 of σ_z respectively.

Next, we construct the projection operators for these eigenstates by taking the outer product of each eigenvector with its bra:

$$\begin{aligned} P_+ &= |0\rangle \otimes \langle 0| \\ &= |0\rangle \langle 0| \\ &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} \\ P_+ &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} P_- &= |1\rangle \otimes \langle 1| \\ &= |1\rangle \langle 1| \\ &= \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} \\ P_- &= \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

Problem 3. Use Projection Operator to find Measurement probability

Use the projector equation to find the probability of measuring spin up and down when it is applied to the eigenvectors of σ_x .

Hints: you should get 50% for all of them.

Solution

From **Problem 1.**, the eigenvectors of σ_x are

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

“Measuring spin up/down” is a measurement of S_z , so the outcome projectors are the ones derived in **Problem 2.**:

$$P_{\uparrow} = |0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad P_{\downarrow} = |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

By the Born rule, a normalized state $|\psi\rangle$ gives outcome $m \in \{\uparrow, \downarrow\}$ with probability

$$P_{\psi}(m) = \langle \psi | P_m | \psi \rangle$$

Each $|\pm\rangle$ carries the factor $\frac{1}{\sqrt{2}}$, so every $\langle \pm | \dots | \pm \rangle$ contributes $\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2}$.

For $|\psi\rangle = |+\rangle$:

$$\begin{aligned} P_{+}(\uparrow) &= \langle + | P_{\uparrow} | + \rangle \\ &= \frac{1}{2} \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ P_{+}(\uparrow) &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} P_{+}(\downarrow) &= \langle + | P_{\downarrow} | + \rangle \\ &= \frac{1}{2} \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ P_{+}(\downarrow) &= \frac{1}{2} \end{aligned}$$

For $|\psi\rangle = |-\rangle$:

$$\begin{aligned} P_{-}(\uparrow) &= \langle - | P_{\uparrow} | - \rangle \\ &= \frac{1}{2} \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ P_{-}(\uparrow) &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} P_{-}(\downarrow) &= \langle - | P_{\downarrow} | - \rangle \\ &= \frac{1}{2} \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ -1 \end{pmatrix} \\ P_{-}(\downarrow) &= \frac{1}{2} \end{aligned}$$

All four outcomes have probability $\frac{1}{2}$, matching the hint:

$$P_{+}(\uparrow) = P_{+}(\downarrow) = P_{-}(\uparrow) = P_{-}(\downarrow) = \frac{1}{2}.$$

This is expected: $|+\rangle$ and $|-\rangle$ are equal-magnitude superpositions of $|0\rangle$ and $|1\rangle$, so a measurement along z is completely random.

Problem 4. Vector normalization by hand and using python 2

Check **Problem 3.** using co-lab.

Hints:

- Use `np.transpose` for transpose.
- Use `np.dot` for multiplication, including outer product.
- Use `np.linalg.eig` to find eigenvalues.

Solution

Problem 3 gave $P_{\pm}(\uparrow) = P_{\pm}(\downarrow) = \frac{1}{2}$ for both eigenvectors of σ_x . We reproduce it in Co-lab: take $|+\rangle, |-\rangle$ from `np.linalg.eig` (returned unit-normalized), build the spin-up/down projectors $|0\rangle\langle 0|, |1\rangle\langle 1|$ as outer products with `np.dot`, and evaluate $\langle\psi|P_m|\psi\rangle$ using `np.transpose` and `np.dot`.

```

1  sigmax = np.array([[0, 1], [1, 0]])
2  _, eigvecs = np.linalg.eig(sigmax)          # |+>, |-> (unit-normalized)
3  plus, minus = eigvecs[:, [0]], eigvecs[:, [1]]
4
5  zero, one = np.array([[1], [0]]), np.array([[0], [1]])
6  P_up = np.dot(zero, np.transpose(zero))      # |0><0|
7  P_down = np.dot(one, np.transpose(one))      # |1><1|
8
9  for psi in (plus, minus):
10     p_up = np.dot(np.transpose(psi), np.dot(P_up, psi))[0, 0]
11     p_down = np.dot(np.transpose(psi), np.dot(P_down, psi))[0, 0]
12     print(p_up, p_down)

```

The notebook (**Appendix 1**) confirms $\langle+\rangle + \langle-\rangle = 1$ and

$$P_+(\uparrow) = P_+(\downarrow) = P_-(\uparrow) = P_-(\downarrow) = 0.5,$$

matching the hand calculation in **Problem 3.**

Problem 5. Orthonormality of Unitary Matrix Row Vectors

Prove that the row vectors of a unitary matrix are orthonormal.

Solution

Let U be an $n \times n$ unitary matrix, so $UU^\dagger = I$, and write its rows as $\mathbf{r}_1, \dots, \mathbf{r}_n$. The (i, j) entry of $UU^\dagger$ is row i of U times column j of $U^\dagger$:

$$(UU^\dagger)_{ij} = \sum_{k=1}^n U_{ik} \overline{U_{jk}} = \sum_{k=1}^n \overline{(\mathbf{r}_j)_k} (\mathbf{r}_i)_k = \langle \mathbf{r}_j | \mathbf{r}_i \rangle,$$

i.e. the Hermitian inner product of rows i and j .

Since U is unitary, $(UU^\dagger)_{ij} = \delta_{ij}$, hence $\langle \mathbf{r}_i | \mathbf{r}_j \rangle = \delta_{ij}$ for all i, j :

- $i = j$: $\langle \mathbf{r}_i | \mathbf{r}_i \rangle = \|\mathbf{r}_i\|^2 = 1$, so every row is a unit vector;
- $i \neq j$: $\langle \mathbf{r}_i | \mathbf{r}_j \rangle = 0$, so distinct rows are orthogonal.

Therefore the row vectors of U form an orthonormal set. (Repeating the argument with $U^\dagger U = I$ shows the columns are orthonormal as well.) ■

Problem 6. Finding the Determinants of Pauli Matrices by Hand

Find the determinants of σ_x , σ_y , and σ_z . What do you notice?

Solution

For a 2×2 matrix, $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$. With

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

this gives

$$\det \sigma_x = (0)(0) - (1)(1) = -1,$$

$$\det \sigma_y = (0)(0) - (-i)(i) = -(-i^2) = -1,$$

$$\det \sigma_z = (1)(-1) - (0)(0) = -1.$$

What we notice: all three Pauli matrices have $\det \sigma_i = -1$. Each σ_i has eigenvalues $+1$ and -1 , and the determinant is the product of the eigenvalues, $(+1)(-1) = -1$, while the trace is their sum, 0 .

Problem 7. Finding the Determinants of Pauli Matrices using Colab

Confirm your answers in **Problem 6**. using Colab. This is a sample code.

```
1 M = np.array([[1, 0], [0, -1]])
2 print(np.linalg.det(M))
```

Solution

Computing each determinant with `np.linalg.det` in Colab:

```
1 sigmax = np.array([[0, 1], [1, 0]])
2 sigmay = np.array([[0, -1j], [1j, 0]])
3 sigmaz = np.array([[1, 0], [0, -1]])
4
5 for name, M in [("x", sigmax), ("y", sigmay), ("z", sigmaz)]:
6     print(name, np.linalg.det(M).real)
```

prints -1 (shown as -1.0000) for all three:

$$\det \sigma_x = \det \sigma_y = \det \sigma_z = -1,$$

confirming the hand calculation in **Problem 6**. For σ_y , `np.linalg.det` returns the complex value $-1 + 0i$, whose real part is -1 . The full notebook is in **Appendix 1**.

Problem 8. Proof of Unitarity

For an orthonormal set of basis, $|v_i\rangle$, we have $\langle v_i | v_j \rangle = \delta_{i,j}$. We transform it using a matrix A , i.e. they become $A|v_i\rangle$. Prove that if after the transformation, they are still orthonormal, then A is unitary.

Hint: Express the basis vectors in column form and each of them has only 1 entry. Express A in its matrix form. The fact that after the transformation, they are still orthonormal implies that the columns of A are orthonormal and thus A is unitary.

Solution

Following the hint, take $\{|v_i\rangle\}$ to be the standard basis, so $|v_i\rangle = \mathbf{e}_i$ is the column vector with a 1 in the i -th entry and 0 elsewhere. For an $n \times n$ matrix A ,

$$A|v_i\rangle = A\mathbf{e}_i = \mathbf{a}_i,$$

the i -th column of A . The hypothesis is that these transformed vectors stay orthonormal:

$$\langle Av_i | Av_j \rangle = \mathbf{a}_i^\dagger \mathbf{a}_j = \delta_{ij}.$$

But $\mathbf{a}_i^\dagger \mathbf{a}_j = \sum_k \overline{A_{ki}} A_{kj} = (A^\dagger A)_{ij}$, so $(A^\dagger A)_{ij} = \delta_{ij}$ for every i, j , i.e.

$$A^\dagger A = I.$$

Since A is square, a left inverse is also a right inverse, so $AA^\dagger = I$ too. Hence A is unitary. ■

The standard basis costs no generality: for *any* orthonormal $\{|v_i\rangle\}$,

$$\langle v_i | (A^\dagger A - I) | v_j \rangle = \langle Av_i | Av_j \rangle - \langle v_i | v_j \rangle = \delta_{ij} - \delta_{ij} = 0$$

for all i, j , and an operator whose every matrix element in a basis vanishes is 0, giving $A^\dagger A = I$ again.

APPENDIX 1

Assignment 04 | Code Solution

assignment-04-code

September 2, 2026

1 Assignment 04

1.1 Import External Libraries

```
[1]: import numpy as np
import sympy as sp

from IPython.display import display, Math
```

1.2 1. Find Eigenvalues using Colab

```
[2]: sigma_x = np.array([[0, 1], [1, 0]])

display(Math(f"\sigma_x = {sp.latex(sp.Matrix(sigma_x))}"))

print("-----")

eigenvalues, eigenvectors = np.linalg.eig(sigma_x)
eigenvalues = np.round(eigenvalues, 4)
eigenvectors = np.round(eigenvectors, 4)

display(Math(f"\text{{Eigenvalues: }} {sp.latex(sp.Matrix(eigenvalues))}"))
display(Math(f"\text{{Eigenvectors: }} {sp.latex(sp.Matrix(eigenvectors))}"))
```

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\text{Eigenvalues: } \begin{bmatrix} 1.0 \\ -1.0 \end{bmatrix}$$

$$\text{Eigenvectors: } \begin{bmatrix} 0.7071 & -0.7071 \\ 0.7071 & 0.7071 \end{bmatrix}$$

1.3 4. Vector normalization by hand and using python 2

```
[3]: # --- Problem 4: check Problem 3 numerically ---
sigma_x = np.array([[0, 1], [1, 0]])

# |+>, |-> are the eigenvectors of sigma_x; np.linalg.eig returns them unit-normalized
_, eigvecs = np.linalg.eig(sigma_x)
plus, minus = eigvecs[:, [0]], eigvecs[:, [1]]

display(Math(f"\ket{{+}} = {sp.latex(sp.Matrix(np.round(plus, 4)))}, \quad "
f"\ket{{-}} = {sp.latex(sp.Matrix(np.round(minus, 4)))}"))
```

```

# normalization check: <psi|psi> = 1 (vectors are real, so transpose = conjugate_
↳ transpose)
for name, psi in [( "+", plus), ("-", minus)]:
    print(f"<{name}|{name}> = {np.dot(np.transpose(psi), psi)[0, 0]:.4f}")

print("-----")

# spin up / down projectors from Problem 2: |0><0| and |1><1| (outer product via np.
↳ dot)
zero, one = np.array([[1], [0]]), np.array([[0], [1]])
P_up = np.dot(zero, np.transpose(zero))
P_down = np.dot(one, np.transpose(one))

# Born rule: P_psi(m) = <psi| P_m |psi>
for name, psi in [( "+", plus), ("-", minus)]:
    p_up = np.dot(np.transpose(psi), np.dot(P_up, psi))[0, 0]
    p_down = np.dot(np.transpose(psi), np.dot(P_down, psi))[0, 0]
    display(Math(f"\text{{{P}}}_{{{name}}}(\uparrow) = {p_up:.4f}, \quad "
f"\text{{{P}}}_{{{name}}}(\downarrow) = {p_down:.4f}"))

```

$$|+\rangle = \begin{bmatrix} 0.7071 \\ 0.7071 \end{bmatrix}, \quad |-\rangle = \begin{bmatrix} -0.7071 \\ 0.7071 \end{bmatrix}$$

$$\langle + | + \rangle = 1.0000$$

$$\langle - | - \rangle = 1.0000$$

$$P_+(\uparrow) = 0.5000, \quad P_+(\downarrow) = 0.5000$$

$$P_-(\uparrow) = 0.5000, \quad P_-(\downarrow) = 0.5000$$

1.4 7. Finding the Determinants of Pauli Matrices using Colab

```

[4]: # --- Problem 7: determinants of the Pauli matrices ---
sigmax = np.array([[0, 1], [1, 0]])
sigmay = np.array([[0, -1j], [1j, 0]])
sigmaz = np.array([[1, 0], [0, -1]])

for name, M in [("\sigma_x", sigmax), ("\sigma_y", sigmay), ("\sigma_z", sigmaz)]:
    det = np.linalg.det(M)
    display(Math(f"\det {name} = {det.real:.4f}"))

```

$$\det \sigma_x = -1.0000$$

$$\det \sigma_y = -1.0000$$

$$\det \sigma_z = -1.0000$$